

Designing Security Controls



Antivirus & EDR (Endpoint Detection and Response)

1. Antivirus Software
2. Antivirus Architecture
3. Antivirus Engine
4. Malware Signatures
5. Endpoint Detection and Response

Labs:

- Students will analyze email headers to find red flags
- students will perform a scan with a given tool from Symantec
- Students will analyze the logs provided to them with the help of the tool that we provide to them

Antivirus vs EDR

- **Antivirus**

- It's a software designed to detect, prevent and remove malicious software.
- AV focuses primarily on known threats.
- Modern AV includes ML-based detection and exploit mitigation.

- **Endpoint Detection and Response (EDR)**

- EDR goes beyond antivirus by providing detection, investigation, and response capabilities.
- EDR is essential for detecting:
 - Lateral movement
 - Living-off-the-land attacks
 - Fileless malware
 - Credential dumping



Antivirus Software

- Antivirus software is a host-based security solution that:
 - ✓ Scans files, memory, and processes
 - ✓ Detects known and unknown malware
 - ✓ Blocks malicious execution in real time
 - ✓ Performs remediation or quarantine
 - ✓ Generates security alerts and logs
- Originally signature-based, modern antivirus solutions now include behavioral analysis, heuristic detection, and cloud intelligence.



Core Functions of Antivirus Software

Malware Detection

- Identifies malicious code using:
 - Known signatures
 - Behavioral patterns
 - Heuristic analysis

Real-Time Protection Monitors:

- File creation and execution
- Process behavior, Memory activity
- Registry changes

On-Demand and Scheduled Scanning

- Full system scans
- Custom path scans
- Removable media scanning

Malware Removal and Quarantine

- Blocks execution
- Isolates malicious files
- Attempts safe remediation

Reporting and Alerting

- Local event logs
- Centralized console reporting
- SIEM integration

Types of Antivirus Detection Techniques

- **Signature-Based Detection**

- Compares files against known malware fingerprints.
- Highly accurate for known threats.
- Ineffective against zero-day malware.

- **Heuristic Analysis**

- Detects suspicious code patterns and structures.
- Identifies variants of known malware.
- Potential false positives.



Types of Antivirus Detection Techniques

- **Behavior-Based Detection**

- Monitors runtime behavior.
- Credential dumping
- Process injection
- Ransomware encryption behavior

- **Cloud-Based Detection**

- Uses threat intelligence and reputation services.
- Faster updates
- Global threat visibility

- **Machine Learning Detection**

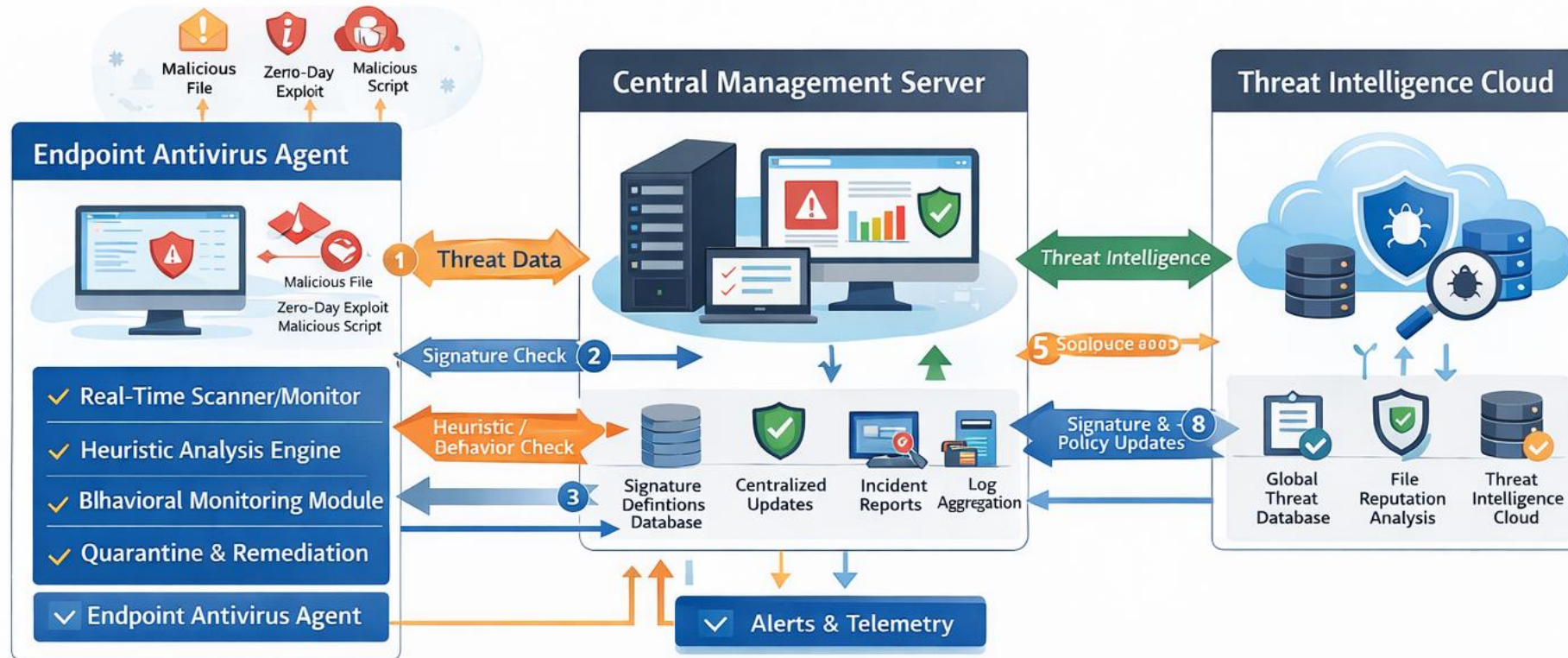
- Uses models trained on malware behavior.
- Improves detection of unknown threats.

Antivirus Architecture

- Antivirus (AV) architecture defines how *malware detection, prevention, analysis, and response* are implemented across endpoints and centrally managed environments.
- Step-by-Step Flow of Antivirus:
 - File is downloaded or executed
 - Endpoint agent intercepts the event
 - Signature and heuristic checks performed
 - Behavior monitored (if allowed to execute)
 - Cloud reputation queried (optional)
 - Verdict returned:
 - Allow
 - Block
 - Quarantine
 - Event logged and reported to console



Antivirus working:



- | | | | |
|--------------------------------------|---------------------------|---------------------------------|----------------------------|
| 1 User Action / File Download | 2 Signature Check | 7 Signature & Policy Updates | 8 Threat Logs & Event Data |
| 3 Signature-Based Detection Engine | Heuristic Analysis Engine | 4 Reputation & Cloud Check | 9 Threat Intelligence |
| 6 Quarantine & Remediation Component | | 8 Logging & Telemetry Component | |



Enterprise Antivirus Deployment Models

- **Standalone Architecture**

- Single endpoint
- No centralized control
- Limited scalability

- **Centralized Enterprise Architecture**

- Endpoint agents + management server
- Suitable for corporate networks

- **Cloud-Managed Architecture**

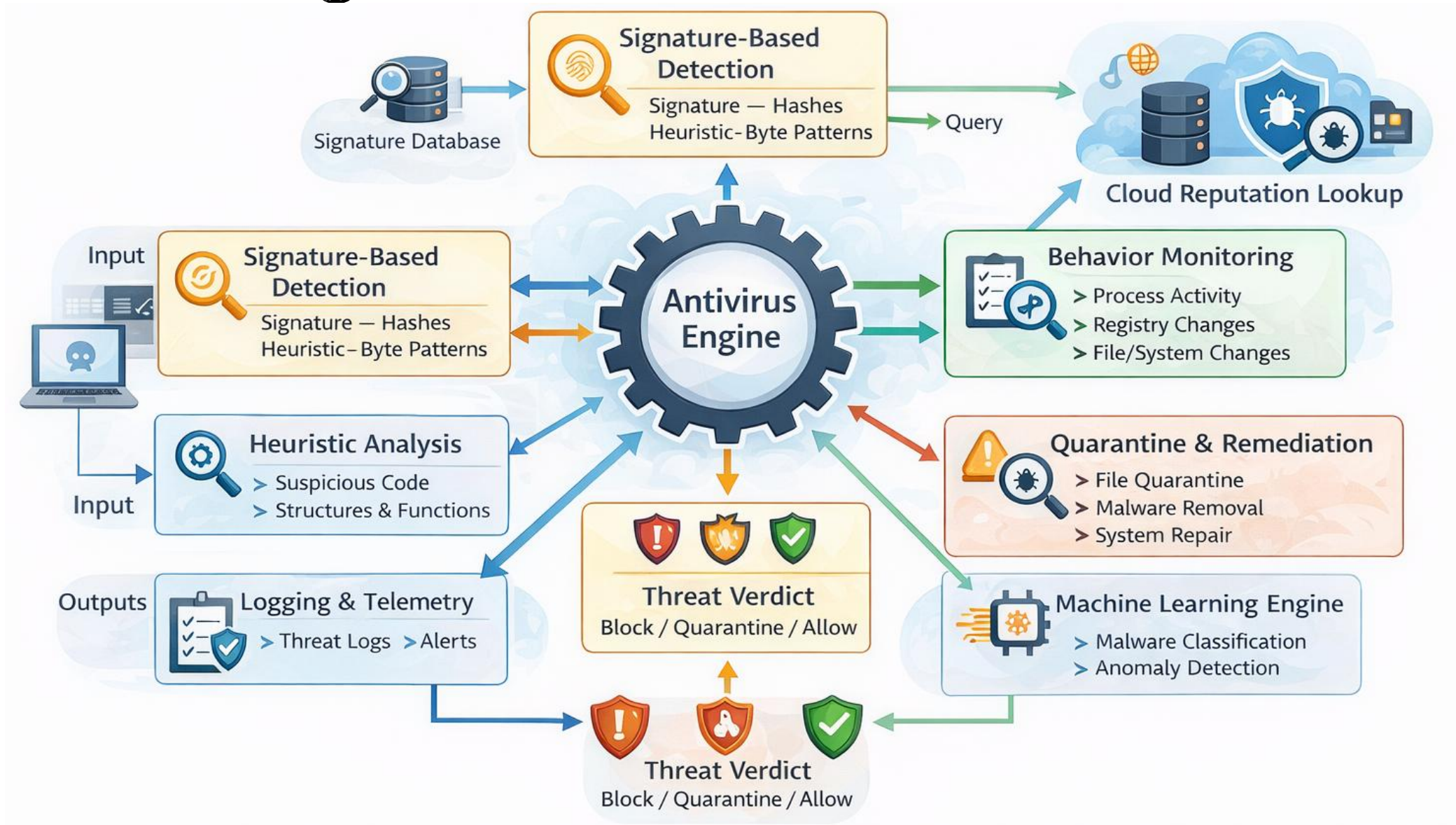
- Lightweight agents
- Vendor-hosted console
- Ideal for remote and hybrid workforces

Antivirus Engine

- An Antivirus Engine is the core analytical and enforcement component of antivirus software responsible for *detecting, classifying, blocking, and remediating* malware.
- It operates at both kernel and user space, continuously analyzing files, processes, memory, and system behavior.
- The antivirus engine is a collection of tightly integrated detection and response modules that:
 - Inspect files before and during execution
 - Analyze code structure and behavior
 - Correlate threat intelligence
 - Enforce security actions (block, quarantine, remediate)
- It functions as the decision-making brain of the antivirus system.



Antivirus Engine



Antivirus Engine vs EDR Engine

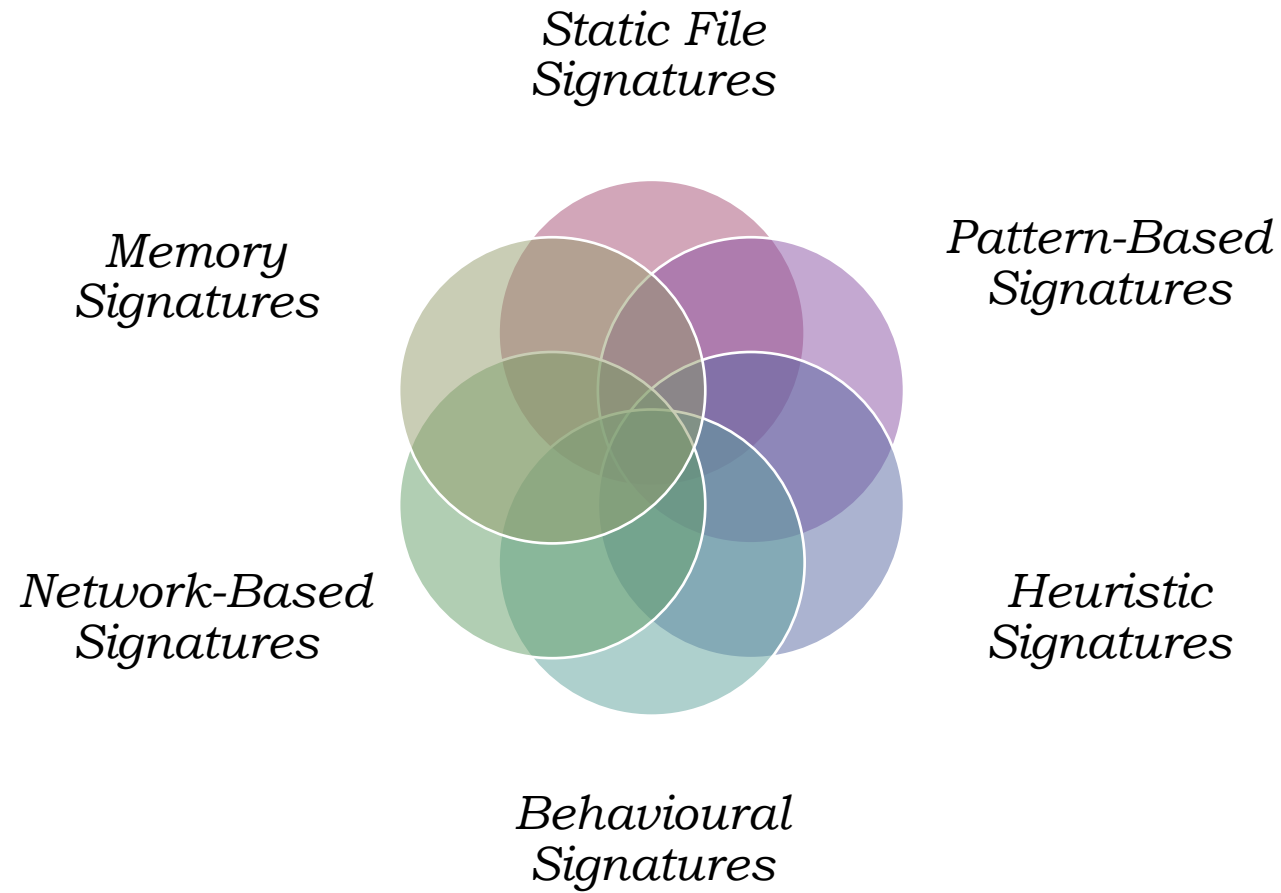
Capability	Antivirus Engine	EDR Engine
<i>Primary role</i>	Prevention	Detection & Response
<i>Behavioural depth</i>	Moderate	Deep
<i>Investigation tools</i>	Limited	Extensive
<i>Lateral movement detection</i>	Weak	Strong

Malware Signatures

- Malware signatures are distinctive patterns or characteristics used by security tools—primarily antivirus and intrusion detection systems—to identify known malicious software.
- They form the foundation of signature-based detection, which remains one of the most reliable and fastest malware identification methods.
- A malware signature is a unique identifier derived from:
 - Malware code
 - File structure
 - Behavioral traits
 - Network indicators
- When a scanned object matches a known signature, it is classified as *malicious*.



Types of Malware Signatures

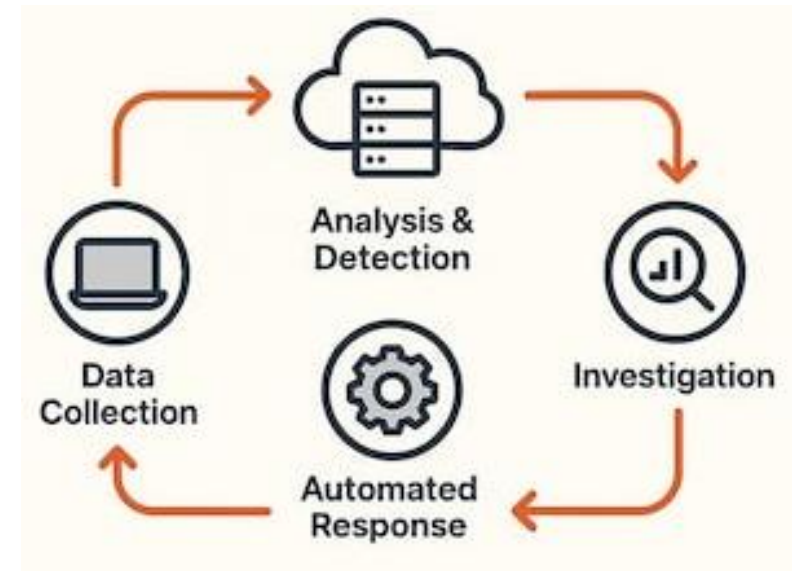


How Malware Signatures Are Created?

- Malware sample collected
- Reverse engineering performed
- Unique characteristics identified
- Signature written and tested
- False positives eliminated
- Signature published to customers

Endpoint Detection and Response (EDR)

- EDR is an advanced endpoint security capability focused on **continuous monitoring, threat detection, investigation, and response for endpoints** such as laptops, servers, and virtual machines.
- EDR addresses the gaps left by traditional antivirus by detecting behavior-based, fileless, and advanced attacks.
- EDR is a security solution that:
 - ✓ Continuously collects endpoint telemetry
 - ✓ Detects malicious and suspicious behavior
 - ✓ Enables threat investigation and hunting
 - ✓ Supports rapid containment and remediation



Why EDR Is Required (Beyond Antivirus)

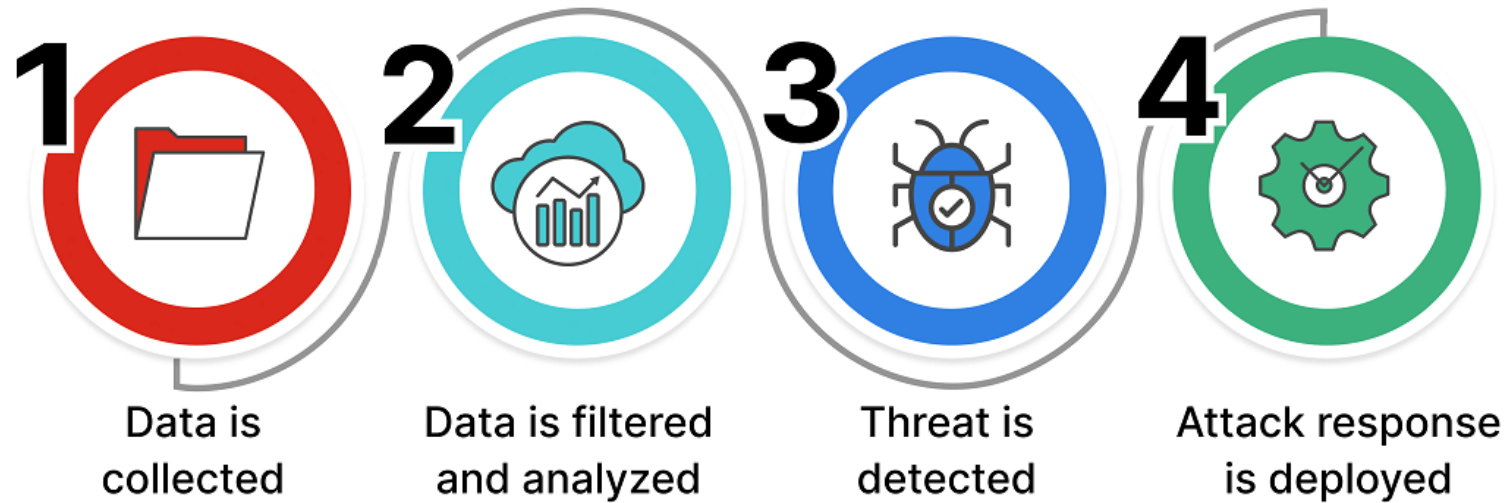
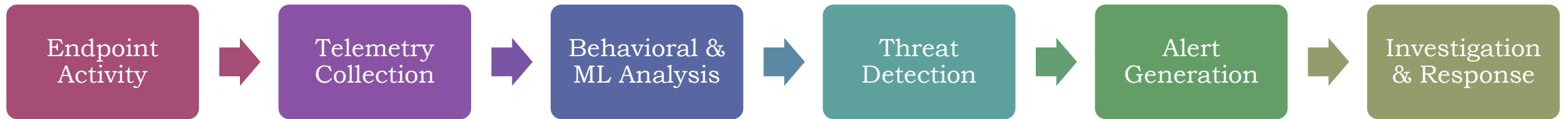
Traditional Antivirus	EDR
Signature-based	Behaviour-based
Preventive	Detective + responsive
Limited visibility	Deep endpoint telemetry
No investigation	Full attack timeline

Core Components of an EDR Solution

- **Endpoint Sensor / Agent** - Collects process creation, Installed on each endpoint
- **Telemetry Collection & Storage** - Retains historical data for investigation
- **Detection Engine** - Behavioral analytics, Threat intelligence
- **Investigation & Threat Hunting Console** - Provides process trees, attack timelines
- **Response & Containment Module** - Kill malicious processes, Quarantine files



EDR Detection Workflow

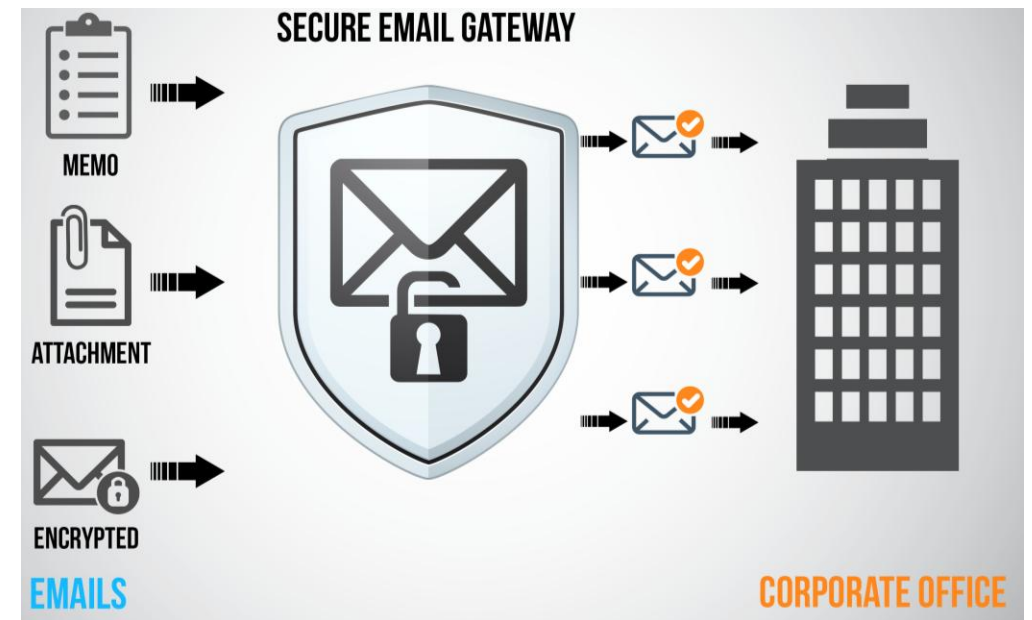


Email Protection

1. Need for email protection
2. Role of email protection
3. What are the options?
4. Email security and backup methods
5. Email protection mechanisms
6. How to fix SPAM emails
7. Best Practices

Need for email protection

- Email remains the primary attack vector for cyber threats because it is ubiquitous, trusted by users, and directly connected to identity and business workflows.
- Effective email protection is therefore mission-critical for preventing breaches, data loss, and financial fraud.
- Industry reality - More than 90% of initial enterprise compromises originate from email.



Why Email Is the Most Targeted Attack Surface?

- Used by every employee, vendor, and customer
- Directly tied to identity (credentials, MFA, OAuth)
- Bypasses perimeter defenses
- Relies on human interaction (clicks, replies, trust)

Major Threats Delivered via Email

- **Phishing** - Fake login pages
- **Business Email Compromise (BEC)** - High financial loss, minimal malware footprint.
- **Malware Delivery** - Ransomware, Trojans, Fileless malware
- **Zero-Day and Advanced Threats** - Weaponized documents
- **Spam and Email Bombing** - Resource exhaustion

Role of email protection

- *Preventing Initial Access Attacks*
 - ✓ Phishing and spear-phishing
 - ✓ Malicious attachments and URLs
- *Protecting Identity and Credentials*
 - ✓ Blocking credential harvesting emails
 - ✓ Detecting brand and domain impersonation
- *Reducing Malware and Ransomware Risk*
 - ✓ Scans attachments using sandboxing and ML
 - ✓ Blocks ransomware delivery at the earliest stage



Role of email protection

- *Detecting and Stopping Business Email Compromise (BEC)*
 - ✓ Analyze sender behavior and writing style
 - ✓ Identify executive impersonation
- *Enabling Post-Delivery Detection and Response*
 - ✓ Continuously re-analyzes delivered emails
 - ✓ Auto-remediates threats retroactively
- *Supporting Security Awareness and Policy Enforcement*
 - ✓ Inserts warning banners for suspicious emails
 - ✓ Supports phishing simulations and reporting



Role of email protection

- *Enhancing Visibility and Threat Intelligence*
 - ✓ Attack trend visibility
 - ✓ Threat intelligence enrichment
- *Ensuring Compliance and Governance*
 - ✓ Data protection regulations (GDPR, HIPAA)
 - ✓ Industry standards (ISO 27001, SOC 2)
- *Integrating with the Broader Security Stack*
 - ✓ IAM and MFA
 - ✓ EDR/XDR



What are the options?

Native Email Security (Built-in Controls)

- Examples
 - Microsoft Defender for Office 365
 - Google Workspace Security
- Capabilities
 - Basic anti-spam and anti-malware
 - Phishing detection
 - Safe links and attachments (limited)
 - SPF, DKIM, DMARC enforcement

Secure Email Gateway (SEG)

- Deployment Model
 - Inline (MX record change)
 - Cloud-based or on-prem
- Capabilities
 - Advanced spam and malware filtering
 - Attachment sandboxing
 - URL rewriting and time-of-click protection
 - Impersonation detection



What are the options?

API-Based Email Security

- Deployment Model
 - API integration with mailbox provider
 - No MX change required
- Capabilities
 - Post-delivery detection
 - Internal email monitoring
 - Advanced BEC and account takeover detection
 - Retroactive email removal

Hybrid Email Protection (SEG + API)

- Architecture
 - SEG for pre-delivery filtering
 - API-based tool for post-delivery detection
- Capabilities
 - Defense-in-depth
 - Maximum phishing and BEC protection
 - End-to-end email visibility

Email Security and Backup Methods

▪ Security mechanisms

- Spam filtering
- Malware scanning
- URL rewriting
- Attachment sandboxing
- Domain authentication
- DLP controls
- Encryption (S/MIME, TLS)

▪ Backup methods

- Cloud backup (O365, Google Workspace backup)
- Email journaling
- Immutable storage
- Retention policies
- Disaster recovery replication

Email security methods

Sender Authentication and transport security

- TLS Encryption – Secures email in transit.
- Sender policy framework – Prevents spoofing

Anti-Spam and Anti-Phishing Controls

- Brand and domain impersonation detection
- Display-name spoofing protection

Malware and Attachment Protection

- Signature-based scanning
- Blocking macros, scripts, and HTML smuggling

URL and Link Protection

- URL rewriting
- Time-of-click inspection

Data Loss Prevention (DLP)

- Policy-based blocking or encryption
- Prevents accidental or malicious data leaks

Business Email Compromise (BEC) Protection

- Behavioral analysis
- Payment and invoice fraud detection

Email backup methods

Native Email Platform Retention

- Microsoft 365 retention policies
- Google Vault

Third-Party Cloud Email Backup

- Automated mailbox backup
- Granular restore

On-Premise Backup (Legacy Systems)

- Mailbox database backup
- File-level backup

Immutable / Air-Gapped Backups

- Write-once storage
- Ransomware-resistant

Backup Scope

- Email backup should include:
 - Mailboxes
 - Attachments
 - Calendars
 - Contacts
 - Shared mailboxes
 - Public folders

Email protection mechanisms

Sender Authentication Mechanisms

- SPF (Sender Policy Framework) - Prevents domain spoofing
- DKIM (DomainKeys Identified Mail) - Digitally signs emails

Transport-Level Security

- TLS Encryption - Encrypts email in transit

Anti-Spam and Anti-Phishing Mechanisms

- Domain similarity and lookalike detection
- Display name spoofing protection



Email protection mechanisms

- *Post-Delivery Protection Mechanisms*
 - Automated mailbox remediation
 - Continuous re-analysis of delivered emails
- *User Awareness and Human-Layer Controls*
 - Phishing warning banners
 - One-click phishing report buttons
- *Email Backup and Recovery (Resilience Mechanism)*
 - Regular mailbox backups
 - Granular restore capability



How to fix SPAM emails?

- Fix Domain-Level Authentication
- Strengthen Email Filtering Controls
- Deploy Advanced Email Security
- Improve Internal Mailbox Hygiene
- Reduce Spam at the User Level
- Control Outbound Email Reputation
- Implement Blocking Policies
- Continuous Monitoring and Tuning



Email Security Best Practices

- **Enforce Email Authentication (Non-Negotiable)**

- *Sender Policy Framework (SPF)* : Authorize only required sending servers
- *DomainKeys Identified Mail (DKIM)*: Enable signing for all outbound mail
- *Domain-based Message Authentication, Reporting, and Conformance (DMARC)*: Move from p=none → quarantine → reject

- **Deploy Layered Email Protection**

- Add Secure Email Gateway (SEG) for pre-delivery filtering
- Add API-based protection for post-delivery detection and BEC

- **Strengthen Attachment and URL Controls**

- Disable Office macros from the internet

Email Security Best Practices

- **Secure Email Accounts (Identity Protection)**
 - Enforce Multi-Factor Authentication (MFA)
- **Enable Post-Delivery Detection and Response**
 - Continuous email re-analysis
- **Implement User Awareness and Reporting**
 - Provide a one-click “Report Phishing” button
- **Maintain Email Backup and Recovery**
 - Use third-party email backup
- **Monitor, Tune, and Audit Continuously**
 - Review spam and phishing metrics monthly

Critical Windows Event IDs

▪ Logon / Logoff

- 4624 — *Successful logon*
- 4625 — *Failed logon*
- 4634 — *Logoff*
- 4647 — User initiated logoff
- 4672 — Special privileges assigned (admin logon)

▪ Credential Abuse Indicators

- 4648 — Logon using explicit credentials
- 4771 — Kerberos pre-auth failed
- 4776 — NTLM authentication attempt
- 4768 — Kerberos TGT requested
- 4769 — Service ticket requested

▪ Account Management & Privilege Escalation

- 4720 — User account created
- 4722 — User account enabled
- 4723 — Attempted password change
- 4724 — Password reset
- 4725 — Account disabled
- 4726 — Account deleted
- 4732 — Added to local group
- 4735 — Security-enabled group modified
- 4740 — Account locked out
- 4767 — Account unlocked

Critical Windows Event IDs

▪ Process Execution & Malware Detection

- 4688 — New process created
- 4689 — Process exited
- 4697 — Service installed
- 7045 (System log) — Service installed (very critical)

▪ PowerShell & Script Monitoring

- 4103 — Module logging
- 4104 — Script block logging
- 4105 — Script execution start
- 4106 — Script execution stop

▪ Persistence Mechanisms

- 4698 — Scheduled task created
- 4702 — Scheduled task updated
- 4719 — System audit policy changed
- 4657 — Registry value modified
- 4663 — File access attempt

▪ Lateral Movement Indicators

- 4624 (Logon Type 3) — Network logon
- 4672 — Admin logon
- 5140 — Network share accessed
- 5145 — Detailed share access
- 4776 — NTLM authentication

Critical Windows Event IDs

▪ **Defender / Antivirus Events**

- 1116 — Malware detected
- 1117 — Malware action taken
- 1118 — Remediation failed
- 1119 — Malware quarantined
- 1121 — Behavior detected

▪ **Ransomware / Suspicious Activity Indicators**

- 4663 — File modifications (mass encryption patterns)
- 4688 — Suspicious execution chain
- 1102 — Security logs cleared
- 4719 — Audit policy change

▪ **Network & Remote Access**

- 5156 — Windows Filtering Platform allowed connection
- 5157 — Blocked connection
- 5158 — Port binding
- 4778 — RDP reconnect
- 4779 — RDP disconnect
- 1149 — RDP logon success

▪ **Email & Identity Related (Exchange / M365 hybrid cases)**

- 4624 — Exchange admin login
- 4648 — Explicit credential usage
- 4688 — Suspicious mail process execution
- 4104 — Email exfil scripts

Tiered Monitoring Priority

■ P1 – Immediate SOC

Alerting

- 4625 spikes
- 4720 / 4728 / 4732
- 7045
- 4688 suspicious processes
- 4104 encoded PowerShell
- 1102 log cleared
- 1116 malware detected
- 4698 scheduled task created

■ P2 – Behavioral

Correlation

- 4624 admin logons
- 5145 share access
- 5156 outbound connections
- 4769 Kerberos anomalies

■ P3 – Threat Hunting

- 4689 process patterns
- 4663 file access trends
- registry modifications

What an Email Header Contains?

- Headers record the technical journey of an email.

- Key components:

- Return-Path
- Received chain
- From / Reply-To
- SPF result
- DKIM signature
- DMARC result
- Message-ID
- X-originating-IP
- MIME & content-type

```
Delivered-To: jitendrastomar5593@gmail.com
Received: by 2002:a05:6918:998b:b0:427:cf55:21b7 with SMTP id lj11csp2273911ysb;
      Mon, 9 Feb 2026 22:12:13 -0800 (PST)
X-Received: by 2002:a05:690e:4004:b0:64a:dafd:2d3e with SMTP id 956f58d0204a3-64adafd305bmr6l;
      Mon, 09 Feb 2026 22:12:12 -0800 (PST)
ARC-Seal: i=2; a=rsa-sha256; t=1770703932; cv=pass;
      d=google.com; s=arc-20240605;
      b=Bz9h3rYhPlIPSo65Xze90Tb+U+BGt4xcmucGYpsDLA6OuTDDIeebkYLdfpyN1EqDyV
      jOHTX1k16/f510r4L1LK1A096WxrvD50XNmGw+Hj2y0/pak1hduQXkF0AZQ8cacU9AFn
      3aTXmy1tBw32mP8TadJ+FosL+y4nKnf+tZorfeMYNZeMCdfZfNqf1UiX1KiuC5xo0cAK
      LyL11+hVzcrgrvY0k0NYvz9K+VP1EVAfH8xT/Fr+AORSzczjY65wVVgaKxpNsRY7cyP4d
      gyxLR7vfBGUjipqJ6GPy30nJ/TefQZbW18ndwz75MmmpJHhJtC1hge/UMZVNDY2eHPvT
      ifww==
ARC-Message-Signature: i=2; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
      h=mime-version:msip_labels:content-language:accept-language
      :in-reply-to:references:message-id:date:thread-index:thread-topic
      :subject:cc:to:from:dkim-signature;
      bh=IosY+E5wtMlydzS54k6oeAqwPDGsphapcJFUF8Uwoao=;
      fh=UQwGK0fsJvf0hnKVbNRp2wMvajAaGTA+VDB1P/2zSCE=;
      b=ADfaNqPEI+oGAoEgf1XrT8XKh8+jPI6i0BRHmMFU2w1Fcn192b4jDF0m+WDeqqDOsE
      kK1ru9Hk9sPIUYxPohf60feYIdaiuGGM44aQaSoBf/dHhfCke6FMpN5IkvdNKBVT/UiM
      KqTedYE0sC3rps1AYBAXUBeoEaEcjbQOLEssp6hepfLaMjh10/HrV2uT3iRWF82OWJAJ
      GQW6k5mGx8jh1Xly4C/64K6hk0QfJmdx7sd2wy+CTUokcrecOKJ+P3B1yGn0zw+am+tF
      R4f/Ka7ZvvhDNxWs9Iwc+Z049pbAtq486B9hErHJUFiKDvHzif5t/lbkehsXsRMuZaeN
      cUWA==;
      dara=google.com
```



Email Header

 Reply

 Reply all

 Forward

 Delete

 Mark unread from here

 Block "Lakshmi Priya S S"

 Report spam

 Report phishing

 Filter messages like this

 Translate

 Print

 Download message

 Show original

Original Message

Message ID	<PN2PR01MB8765CBC3FB1C2E1C597819EA9A62A@PN2PR01MB8765.INDPRD01.PROD.OUTLOOK.COM>
Created at:	Tue, Feb 10, 2026 at 11:42 AM (Delivered after 4 seconds)
From:	Lakshmi Priya S S <lakshmi.priya@rpsconsulting.in>
To:	Jitendra Singh Tomar <jitendrastomar5593@gmail.com>
Subject:	RE: Use of AI agent by participant on MS Teams
SPF:	PASS with IP 2a01:111:f403:c409:0:0:0:1 Learn more
DKIM:	'PASS' with domain rpsconsulting.onmicrosoft.com Learn more
DMARC:	'PASS' Learn more

[Download Original](#)



Email Header

Original Message

Message ID	<PN2PR01MB8765CBC3FB1C2E1C597819EA9A62A@PN2PR01MB8765.INDPRD01.PROD.OUTLOOK.COM>
Created at:	Tue, Feb 10, 2026 at 11:42 AM (Delivered after 4 seconds)
From:	Lakshmi Priya S S <lakshmi.priya@rpsconsulting.in>
To:	Jitendra Singh Tomar <jitendrastomar5593@gmail.com>
Subject:	RE: Use of AI agent by participant on MS Teams
SPF:	PASS with IP 2a01:111:f403:c409:0:0:0:1 Learn more
DKIM:	'PASS' with domain rpsconsulting.onmicrosoft.com Learn more
DMARC:	'PASS' Learn more

[Download Original](#)

Copy to clipboard

Email Header

```
Delivered-To: jitendrastomar5593@gmail.com
Received: by 2002:a05:6918:998b:b0:427:cf55:21b7 with SMTP id lj11csp2273911ysb;
    Mon, 9 Feb 2026 22:12:13 -0800 (PST)
X-Received: by 2002:a05:690e:4004:b0:64a:dafd:2d3e with SMTP id 956f58d0204a3-64adafd305bmr6647723d50.90.1770703932969;
    Mon, 09 Feb 2026 22:12:12 -0800 (PST)
ARC-Seal: i=2; a=rsa-sha256; t=1770703932; cv=pass;
    d=google.com; s=arc-20240605;
    b=Bz9h3rYhPlIPSo65Xze90Tb+U+BGt4xcmucGYpsDLA6OuTDDIeebkYLdfpyN1EqDyV
    jOHTXlk16/f510r4L1LK1A096WxrvD50XNmGw+Hj2y0/pak1hduQXkF0AZQ8cacU9AFn
    3aTXmy1tBw32mP8TadJ+FosL+y4nKnf+tZorfeMYNZeMCdfZfNqf1UiX1KiuC5xo0cAK
    LyL11+hVzcrGvY0k0NYvz9K+VP1EVAfH8xT/Fr+AORSzczjY65wVVgaKxpNsRY7cyP4d
    gyxLR7vfBGUjipqJ6GPy3OnJ/TefQZbWl8ndwz75MmpJHhJtClhge/UMZVNDY2eHPvT
    ifww==
ARC-Message-Signature: i=2; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
    h=mime-version:msip_labels:content-language:accept-language
    :in-reply-to:references:message-id:date:thread-index:thread-topic
    :subject:cc:to:from:dkim-signature;
    bh=IosY+E5wtMlydzS54k6oeAqwPDGsphapcJFUF8Uwoao=;
    fh=UQwGK0fsJvf0hnKVbNRp2wMvajAaGTA+VDB1P/2zSCE=;
    b=ADfaNqPEI+oGAoEgf1XrT8XKh8+jPI6i0BRHmMFU2w1Fcn192b4jDF0m+WDeqgDOsE
    kK1ru9Hk9sPIUYxPohf60feYIdaiuGGM44aQaSoBf/dHhfCke6FMpN5IkvdNKBVT/UIm
    KqTedYE0sC3rps1AYBAXUBeoEaEcjbQOLEssp6hepfLaMjh10/HrV2uT3iRWf820WJAJ
    GQW6k5mGx8jhlXly4C/64K6hk0QfJmdx7sd2wy+CTUokcrecOKJ+P3B1yGn0zw+am+tF
    R4f/Ka7ZvvhDNxWs9IWc+Z049pbAtq486B9hErHJUFiKDVhziF5t/lbkehsXsRMuZaeN
    cUWA==;
    dara=google.com
ARC-Authentication-Results: i=2; mx.google.com;
    dkim=pass header.i=@rpsconsulting.onmicrosoft.com header.s=selector2-rpsconsulting-onmicrosoft-com header.b=CwGQkCX2;
    arc=pass (i=1 spf=pass spfdomain=rpsconsulting.in dkim=pass dkdomain=rpsconsulting.in dmarc=pass fromdomain=rpsconsulting.in);
    spf=pass (google.com: domain of lakshmi.priya@rpsconsulting.in designates 2a01:111:f403:c409::1 as permitted sender) smtp.mailfrom=lakshmi.pr
    dmarc=pass (p=REJECT sp=REJECT dis=NONE) header.from=rpsconsulting.in
Return-Path: <lakshmi.priya@rpsconsulting.in>
Received: from MA0PR01CU009.outbound.protection.outlook.com (mail.southindia037n1701000001.outbound.protection.outlook.com [2a01:111:f403:c409::1])
```

Header Analyses

Step 1 — Check the “From” vs “Return-Path”

Step 2 — Analyze the “Received” Chain

Step 3 — SPF Authentication

Step 4 — DKIM Signature

Step 5 — DMARC Policy

Step 6 — Message-ID Analysis

Step 7 — Reply-To Manipulation

Step 8 — X-Originating-IP

Step 9 — Attachment & MIME Clues

Quick Start for Symantec Endpoint Protection

- <https://techdocs.broadcom.com/us/en/symantec-security-software/endpoint-security-and-management/endpoint-protection/all/getting-up-and-running-on-for-the-first-time-v45150512-d43e1033/symantec-endpoint-protection-quick-start-guide-v116381733-d25e6.html>





That's all Folks!